

Introduction to Business Analytics

QBA 5330 — Module 1, Part 1 (September 5)

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Where We're Headed Today

- **Part 1:** Introduction to business analytics & framing business questions
- **Part 2:** Sampling and gathering data; observational vs. experimental data
- **Part 3:** Tables, graphs, and descriptive statistics

Quick Check

Show of hands:

- Installed R, RStudio, Excel Solver, and Quarto?
- Read about the analytics decision cycle?
- Practiced importing data and identifying variable types?

The Analytics Decision Cycle

Frame the decision → obtain appropriate data → summarize the evidence → quantify uncertainty → build a model → recommend an action → communicate the result

Today: **Steps 1 and 2** — plus the data vocabulary we need before anything else.

Three Kinds of Analytics

- **Descriptive** — what happened?
 - tables, graphs, summary statistics (*Module 1*)
- **Predictive** — what is likely to happen?
 - regression, forecasting (*Modules 4–5*)
- **Prescriptive** — what should we do about it?
 - optimization (*Module 6*)

What Connects Them

Inferential statistics (confidence intervals, hypothesis tests – Modules 2–3) is the bridge:

- lets us move from a **sample** to justified claims about a **population or process**
- makes prediction and prescription **trustworthy**, not just plausible-looking

Your Turn

Give one example from your own organization of:

1. A **descriptive** analytics tool
2. A **predictive** analytics tool
3. A **prescriptive** analytics tool

A Vague Concern Is Not a Question

“Our customer satisfaction seems to be dropping.”

- This is a **concern**, not something anyone can act on yet.

Framing It as a Decision

“Should we change our current return policy for the upcoming quarter, given a hypothesis that a stricter policy introduced in March is driving dissatisfaction?”

- Names a specific **decision**
- Names a **timeframe**
- Names a candidate **explanation** — tells you what data to collect
- Implies what would count as **evidence** for or against acting

Three Framing Questions

- What **decision** is on the table?
- **Who** is making it, and what are the realistic options?
- What would **change** if the analysis came out one way vs. another?

A Gut-Check

If your analysis wouldn't change anyone's action, it probably wasn't the right analysis to run.

- Apply this to: *“Satisfaction seems to be dropping.”*

Recap: Data Types

- **Categorical** — places observations into named groups
- **Quantitative** — measured on a numeric scale
 - **Discrete** — countable values only
 - **Continuous** — any value within a range

Classify These

An EMBA cohort roster:

Column	Type?
student_id	?
industry	?
years_experience	?
undergrad_gpa	?
completed_capstone (yes/no)	?

Why This Matters

Every summary statistic, every graph, and every model choice later in this course depends on classifying variables correctly **first**.

One More Term: The Observational Unit

What does one row of the dataset represent?

- `store_sales.csv`: one row = one store's performance in one **month** (a “store-month”), not just “a store”
- The EMBA roster: one row = one **student**

Quick Check

A dataset of `date`, `website_visits`, `bounce_rate`, recorded once per day for a year.

What's the observational unit?

Why This Is Worth Slowing Down For

“Average revenue” in `store_sales.csv` is an average **per store-month**.

Not per store. Not per year.

A report that’s vague about this can quietly mislead a reader who assumes otherwise.

Activity: Frame a Business Question

In groups of 3–4, pick one prompt:

1. A regional bank worried about branch staffing costs
2. A hospital system worried about patient no-shows
3. A software company worried about free-trial-to-paid conversion
4. A manufacturer worried about a rise in product returns

Write: a framed decision (not just a topic) + one data type you'd need.

(5 minutes – then we'll share out)

Framing for the Module 1 Deliverable

Worth **10 of 100 points** on the actual rubric.

Weak (a topic):

“This report analyzes store sales data. Sales are an important part of any business...”

Strong (a decision):

“...The company must decide how to allocate next quarter’s inventory and staffing budget across regions. If one region is consistently outperforming the others

Your Turn: One-Sentence Framing (2 minutes)

Write **one sentence** framing your own deliverable dataset as a **decision**, not a topic.

(Not collected — just a first draft, while “strong vs. weak” is still fresh.)

Going Forward

Everything in the next part assumes you've already **framed the decision**.

Sampling and data collection only make sense once you know what question you're answering.